

Domain Model Exercise

1 Question

The system allows users to browse, search, and purchase books. It also handles inventory management, order processing, and customer accounts.

Tasks:

- Identify the key domain entities and attributes.
- Draw a Domain Model diagram to show the relationships between these entities.

1. Key Domain Entities and Attributes

Based on the scenario, the core domain entities and their essential attributes are:

Entity	Attributes (Examples)
Book	ISBN (Primary Key), Title, Author, Publisher, Publication Year, Price, Genre
User / Customer	Customer ID (Primary Key), Name, Email, Password (hashed), Shipping Address, Billing Address
Order	Order ID (Primary Key), Customer ID (Foreign Key), Order Date, Total Amount, Shipping Status (Pending, Shipped, Delivered)
OrderItem	Item ID (Primary Key), Order ID (Foreign Key), ISBN (Foreign Key), Quantity, Unit Price
Inventory	Inventory ID (Primary Key), ISBN (Foreign Key), Stock Quantity, Location

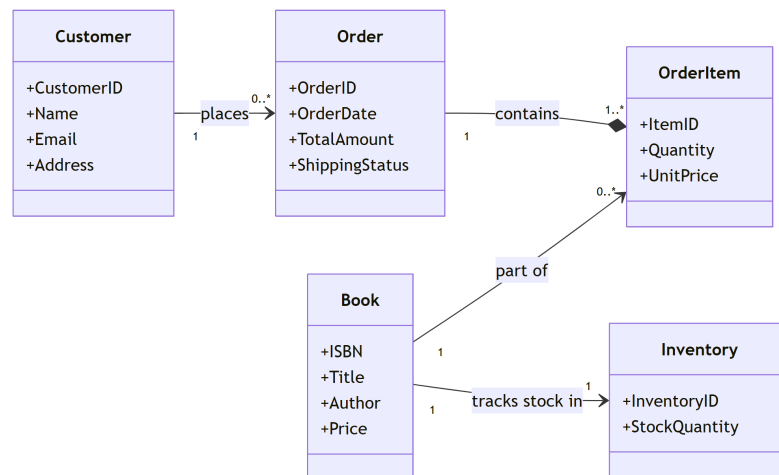
2. Domain Model Diagram (Text Representation)

- Customer** ↔ **Order**: One Customer places zero or more Orders. Each Order is placed by exactly one Customer.
Multiplicity: 1 Customer ↔ 0..* Order
- Order** ↔ **OrderItem**: One Order contains one or more Order Items. Each Order Item belongs to exactly one Order.
Multiplicity: 1 Order ↔ 1..* OrderItem (Composition relationship)
- Book** ↔ **OrderItem**: One Book can be part of zero or more Order Items

across different orders. Each Order Item references exactly one Book.
Multiplicity: 1 Book $\leftrightarrow$ 0..* OrderItem

- **Book $\leftrightarrow$ Inventory**: One Book has exactly one Inventory record (to track its stock). An Inventory record tracks exactly one Book.
Multiplicity: 1 Book $\leftrightarrow$ 1 Inventory

3. Domain Model Diagram



2 Question

A customer visits a store and buys items. The cashier scans the items, calculates the total, receives payment, and prints a receipt. The store keeps track of sales and inventory.

Tasks:

- Identify the key noun phrases and conceptual classes.
- Determine the relationships/associations between the classes.
- Identify attributes for each class.
- Draw a Domain Model diagram showing the connections.

Step 1: Noun Phrases

Customer, Store, Items, Cashier, Total, Payment, Receipt, Sales, Inventory

Step 2: Conceptual Classes

Noun Phrase	Conceptual Class?	Reason
Customer	Yes	Represents a person in the domain
Store	Yes	Legal entity, physical location
Items	Yes	Physical products
Cashier	Yes	Person handling transactions
Total	No (attribute)	Can be calculated from items
Payment	Yes	Transactions are important
Receipt	Maybe	Report of sale; optional until returns scenario
Sales	Yes	Transactions themselves
Inventory	Maybe	Could be modeled as a system or class depending on detail

Step 3: Relationships / Associations

- Sale contains SalesLineItems (line items)
- SalesLineItem represents a specific Item
- Customer makes a Sale
- Cashier processes a Sale
- Store has Cashiers and Items

Step 4: Attributes

Class	Attributes Examples
Customer	name, customerID
Store	name, address
Item	itemID, description, price
Sale	dateTime, total
SalesLineItem	quantity, subtotal
CashPayment	amountTendered
Cashier	cashierID

Step 5: Domain Model Diagram

